

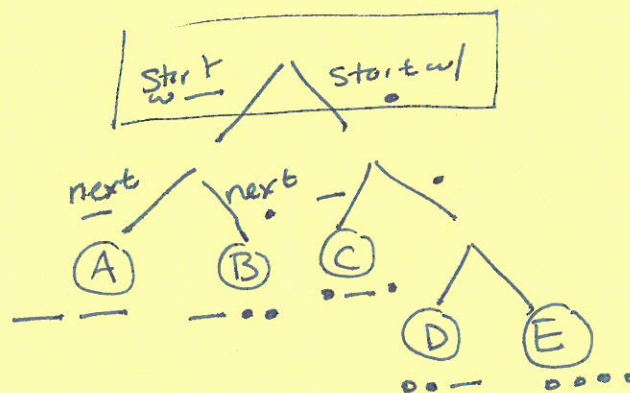
11 October 2023

2 types of notes: $\underset{\uparrow}{-}$ and $\underset{\uparrow}{\cdot}$
 2 beats 1 beat

Q: How many ~~measures~~ ^{measures} are there with n beats?

n		$M(n)$
1	$\cdot$	1
2	$-$ OR $\cdot\cdot$	2
3		
4	$- -$ OR $\cdot\cdot\cdot$ OR $- \cdot\cdot$ OR $\cdot\cdot -$ OR $\cdot - \cdot$	5

one approach:



$M(n) :=$
 total # measures w/ n beats

$=$ # measures that begin w/ $-$ + # measures that begin w/ $\cdot$

$M(n-2) + M(n-1)$

Base cases: $M(0) = 1$ $M(2) = 2$
 $M(1) = 1$

these are our Fibbon. #s (off-by 1 indexing)

$$M(n) = F(n+1)$$

↳ def as:

$$F(0) = 0$$

$$F(1) = 1$$

$$F(n) = F(n-1) + F(n-2) \text{ orw}$$

$$F(n) \in \Theta(\phi^n)$$

↑ the golden ratio

$$\frac{\sqrt{5}+1}{2} \approx 1.618...$$

i.e., this is exponential.

Let's turn our recursive formula into a recursive function

compute num measures (n)

if $n=0$ or $n=1$

| return 1.

else

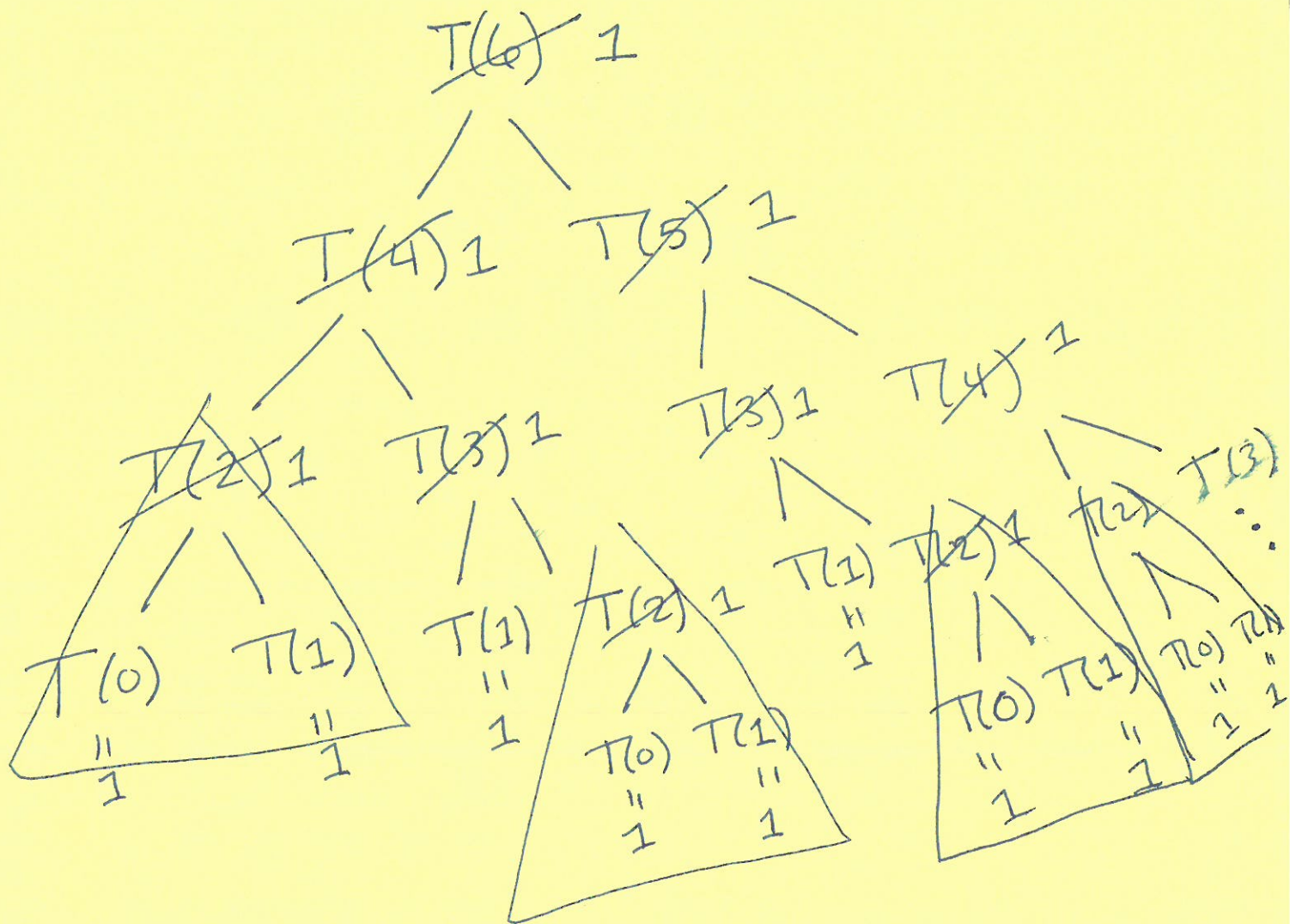
| return compute num meas (n-1) +
compute num meas (n-2)

end else

Recursive formula for worst case time complexity is:

$$T(n) = \Theta(1) + T(n-1) + T(n-2) \in \Theta(\phi^n)$$

For $n=6$, what does our recursion tree look like?



Same subtree happens again and again...

Fix 1 : "Simplest" approach to DP
(as-in, the least change to
your existing algorithm)
Save the results!
"memoization"

note: need some DS to store this in.

// M is some DS storing our results
CNM2(n)

if $n=0$ or $n=1$
return 1

else if $M(n)$ not known

$M(n) \leftarrow \text{~~1~~ (NM2(n-1) + CNM2(n-2))}$

end else

end if

Q: How many additions
(before: was exponential)
happen now?

n	0	1	2	3	4
# adds	0	0	1	2	3

additions is $n-1 \in \Theta(n)$

Fix 2: FILL DELIBERATELY (what most folks think of as DP)

- ① Figure out a good DS
- ② Determine nice computation order
so you never try to access a value
before needed

$$\underbrace{T(n)}_{\text{compute 2nd}} = \underbrace{T(n-1)}_{\text{compute 1st}} + T(n-2)$$

CNM3(n)

if $n=0$ or $n=1$

| return 1

end if

~~values~~

~~array~~

values $\leftarrow$ array of length n
s.t. $\text{values}[0]$ and $\text{values}[1] = 1$

for $k=3$ to n

| $\text{values}[k] \leftarrow \text{values}[k-1] + \text{values}[k-2]$

end for

return $\text{values}[k]$